

Swiss Institute of Banking and Finance



University of St.Gallen

4. Corporate Valuation

Prof. Dr. Markus Schmid
3,130/4,130 Corporate Finance

Course outline

1. General Information
2. Risk and Return
3. Portfolio Theory and CAPM
- 4. Corporate Valuation**
5. Cost of Capital
6. Corporate Financing and the Issuance of Securities
7. Capital Structure
8. Risk Management
9. Conflicts of Interest and Corporate Governance
10. Market Efficiency and Behavioral (Corporate) Finance
11. Empirical Research in Corporate Finance

Agenda

1. Example stock: VAT Group AG (VACN)
2. Equity vs. Enterprise Value
3. Dividend Discount Model
4. Gordon Growth Model
5. Discounted Cash Flow Model
6. Multiple Valuation

VAT Group AG – Company information

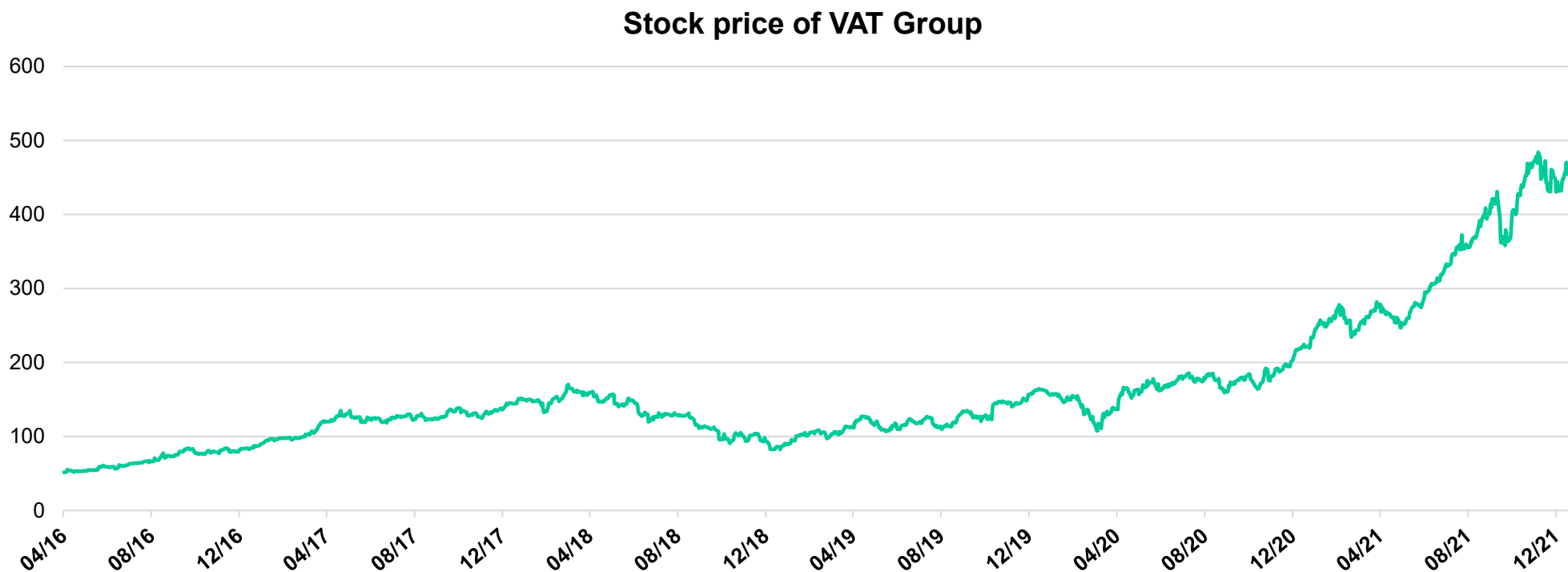
- **VAT Group AG** (proper spelling "VAT Vakuumventile AG") is an internationally active Swiss company for vacuum valves and related services in the semiconductor, display and solar industries as well as in the industrial and research sectors, headquartered in Haag in the municipality of Sennwald in the canton of St. Gallen (Switzerland). Among others, the company's products are used in semiconductor and flat panel display manufacturing, glass and tool coating, metallurgy, surface analysis, high-energy physics, synchrotrons, laser technology and space simulation. VAT Group is listed on the SIX Swiss stock exchange (ticker symbol: VACN)
- As of year-end 2021, VAT Group generated revenues of CHF 901.2m and an EBIT of CHF 264.9m.
- Total assets (= the book value of the company) are CHF 1064.9m, made up by CHF 430.5m liabilities and CHF 634.4m equity.

VAT Group AG – Market information

Valor symbol	VACN
ISIN	CH0311864901
IPO date	14.04.2016
IPO Issue Price	CHF 45
First-day Closing Price	CHF 51.55
Number of shares outstanding	30,000,000
Share Price (31.12.2021)	CHF 454.40
EPS	CHF 7.25
P/E-Ratio	62.7
Dividend 2021	CHF 4.50
Market Capitalization (31.12.2021)	CHF 1.36bn

VAT Group AG – Market information (II)

- VAT Group's stock price development from April 16, 2016, to December 31, 2021:



Source: Datastream

VAT Group AG – Valuation example

- So what is the value of VAT Group AG as of 31 December 2021?
- What purpose serve valuation tools then for valuing / evaluating VAT Group AG?

Equity vs. enterprise value

All equity financed firm

Value of assets	Value of equity
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- If a company is completely financed with equity, the value of equity equals the value of assets.

Partially equity financed firm

Value of assets	Value of liabilities
	Value of equity

- If a company is financed with equity and debt, the sum of the value of equity and the value of debt equals the value of assets.

Dividend Discount Model

Idea: From a stock investment (e.g., from $t = 0$ to $t = T$), a stockholder can expect to receive all future dividends (D_t) plus the price (P_T) when selling the stock.

$$PV_0 = \frac{D_1}{(1+r)^1} + \frac{D_2}{(1+r)^2} + \dots + \frac{D_T + P_T}{(1+r)^T}$$

$$PV_0 = \sum_{t=1}^T \frac{D_t}{(1+r)^t} + \frac{P_T}{(1+r)^T}$$

$$PV_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t} \quad \text{with} \quad \lim_{t \rightarrow \infty} \frac{P_t}{(1+r)^t} = 0$$

Dividend Discount Model – Example

Company XYZ will pay dividends of CHF 3, CHF 3.24, and CHF 3.50 over next three years. After three years, the stock sells for CHF 94.48.

What is the stock price given an expected return (r) of 12%?

$$PV_0 = \frac{3.00}{(1 + 0.12)^1} + \frac{3.24}{(1 + 0.12)^2} + \frac{3.50 + 94.48}{(1 + 0.12)^3} = 75.00$$

In practice, it is difficult to estimate stock prices in this framework as we have to estimate future dividends (in principle until infinity) and the appropriate discount rate (i.e., expected return).

Gordon Growth Model

- The Gordon Growth Model (GGM) is a special case of the Dividend Discount Model (DDM).
- The equity value is determined by dividend payments (D) to shareholders, assuming a constant dividend growth (g) and $r > g$:

$$\text{Equity value} = \frac{D_1}{1+r} + \frac{D_1 \cdot (1+g)}{(1+r)^2} + \frac{D_1 \cdot (1+g)^2}{(1+r)^3} + \dots = \frac{D_1}{r-g}$$

- This valuation formula is often used for simplicity but implies a **constant dividend growth rate to infinity** (or a very long period of time)!

Gordon Growth Model – Real world example Con Ed

Consolidated Edison

- Quasi-monopolistic power supplier of New York City
- Demand for utilities is stable since decades
- Prices are set by the authorities to meet inflation rates
- Stable leverage ratios since decades
- The firm pays out as much dividends as possible (95% of the free cash flow during 2006 – 2010)
- Dividend per share was \$2.22 in 2010
- The risk free rate was 3.5%.
- The market risk premium is expected to be 5%.
- Utilities face an average equity beta of 0.8.
- Dividends can be assumed to grow with the risk free rate.

Source: A. Damodaran (2012), Investment Valuation, p. 326 f.

Gordon Growth Model – Real world example Con Ed (II)

What is Consolidated Edison's share price based on a valuation with the Gordon Growth Model?

1. Calculate the cost of equity (under the CAPM):

2. Calculate the share price:

Gordon Growth Model – Real world example VAT Group

What is VAT Group's share price based on a valuation with the Gordon Growth Model as of 31.12.2021?

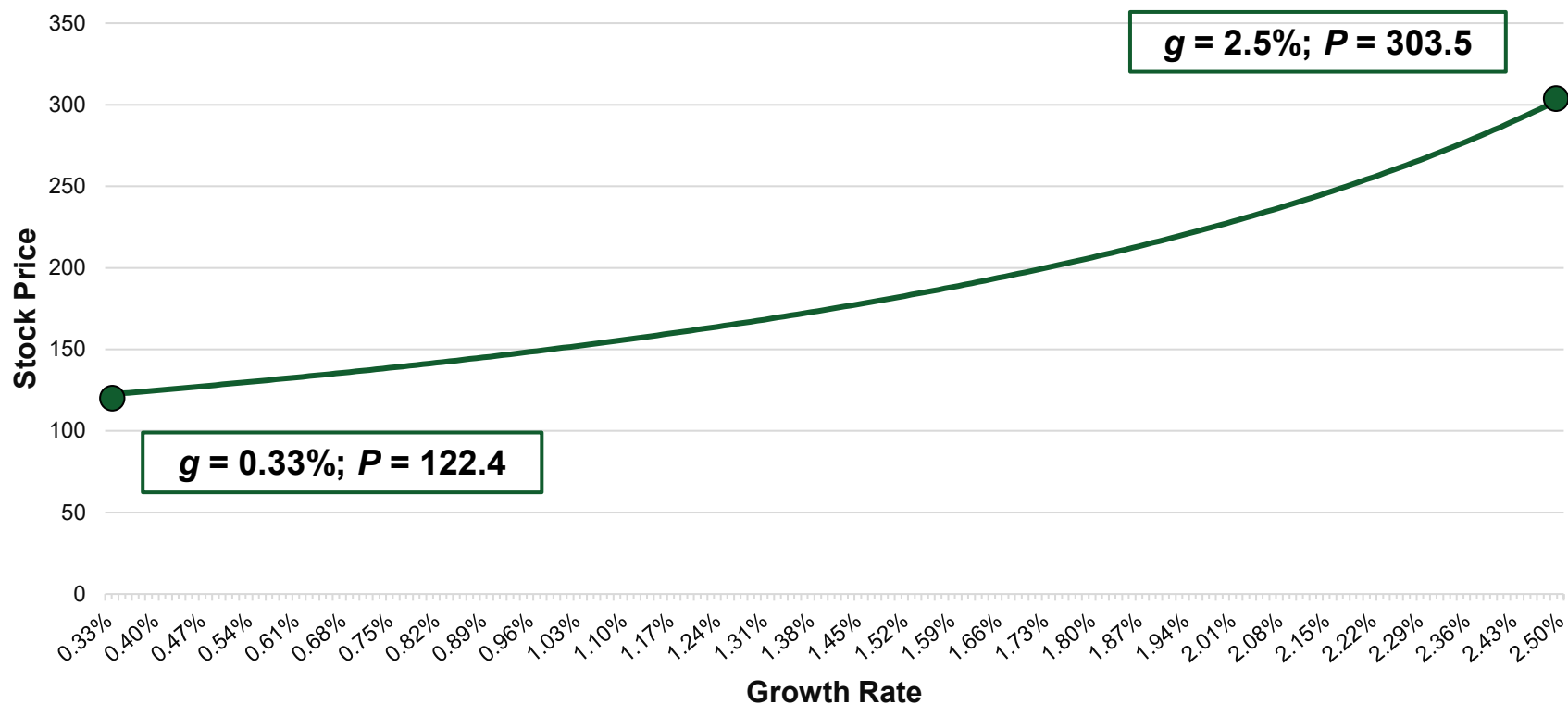
1. The cost of equity (under the CAPM) is estimated to be:
2. Calculate the share price:
 $Div_0 = \text{CHF } 4.50$; $g = ? \rightarrow$ assumption: $g = 0.33\%$ (long-term average risk-free rate from SNB)

Gordon Growth Model – Real world example VAT Group (II)

- In case of VAT Group, the price is clearly off. Is our estimate of g inadequate? Or is the market wrong?
- What is the growth rate implied by the observed share price?

Gordon Growth Model – Real world example VAT Group (III)

- How sensitive is VAT Group's share price to variations in the constant growth rate, g ?



Source: Datastream

Gordon Growth Model – Estimating growth rates

Idea: If a firm pays out all its earnings, it will not grow. If the firm retains some of its earnings, these earnings will be reinvested and impact the growth rate of the firm.

$$g = \underbrace{ROE}_{\text{Return on Equity}} \cdot \underbrace{\text{plowback ratio}}_{\text{Retained earnings}}$$

$$ROE = \frac{EPS}{\text{Book Value per Share}}$$

$$\text{Plowback ratio} = 1 - \frac{\text{Dividend}}{EPS}$$

As useful as the constant-growth DDM is, remember that it is based on a (very!) simplifying assumption. In fact, firms pass through **life cycles with different dividend profiles**.

Gordon Growth Model & growth opportunities

A company plans to pay a \$8.33 dividend for infinity (100% of earnings). The expected return on equity (r) is 15%. Instead of paying out \$8.33, the company can plow back 40% of its earnings at the firm's return on equity (ROE) of 25%.

What is the stock value before and after the plowback decision?

Value without growth

$$P_0 = \frac{8.33}{0.15} = 55.53$$

Value with growth

$$\begin{aligned} g &= 0.25 \cdot 0.40 = 0.1 \\ D_1 &= 8.33 \cdot (1 - 0.4) = 5 \\ P_0 &= \frac{5}{0.15 - 0.1} = 100 \end{aligned}$$

Gordon Growth Model & growth opportunities (II)

- The difference is **called present value of growth opportunities (PVGO)**:

Value with growth		Value without growth		PV of growth opportunities
$P_0 = \frac{5}{0.05} = 100$	—	$P_0 = \frac{8.33}{0.15} = 55.53$	=	$PVGO = 44.47$

- Interpretation: The value of a growth stock is much higher than the value of a non-growth stock. Remember that a growth firm needs to have profitable projects to invest in ($ROE > r$).

Gordon Growth Model & growth opportunities (III)

- PVGO is the net present value of a firm's future investments.
- The increase in the stock price reflects the fact that the planned investments provide an expected rate of return greater than the required rate. In other words, the **investment opportunities** have a **positive NPV**.
- Growth per se is not what investors desire; growth enhances firm value only if it is achieved by investing in projects with attractive profit opportunities.
- Most investors prefer a steady growth rate at which a firm can grow. The so-called **sustainable growth rate** is the highest growth rate that the firm can maintain without increasing its financial leverage (see lecture 7).

Payout ratio

- The payout ratio is the proportion of income which is paid out to shareholders.
- Payout policy resolves two questions:
 - (1) How much cash should the corporation pay out to its shareholders?
 - (2) How should the cash be distributed, by paying cash dividends or repurchasing shares?
- Dividend policy:

High payout ratio

High dividend yield today, but limited growth implies (relatively) lower dividend payments in the future.

Low payout ratio

Low dividend yield today, but strong growth implies (relatively) higher dividend payments in the future.



Problems of DDM & Gordon Growth Models

- The dividend discount model is a very intuitive and straightforward model.
- However, young and high-growth firms often do not pay dividends.
- Moreover, dividends are the strictest / narrowest measure of cash flow to equity and many firms **do not pay dividends**. In fact, the number of dividend payers decreased (e.g., Fama and French, 2001; von Eije and Megginson, 2008).
- In contrast, more firms **buy back shares** to return cash to their shareholders. In 2007, the total value of share buybacks was about double the value of stock dividends in the US.
- Concentrating on dividends would result in an erroneous valuation.

Problems of DDM & Gordon Growth Models (II)

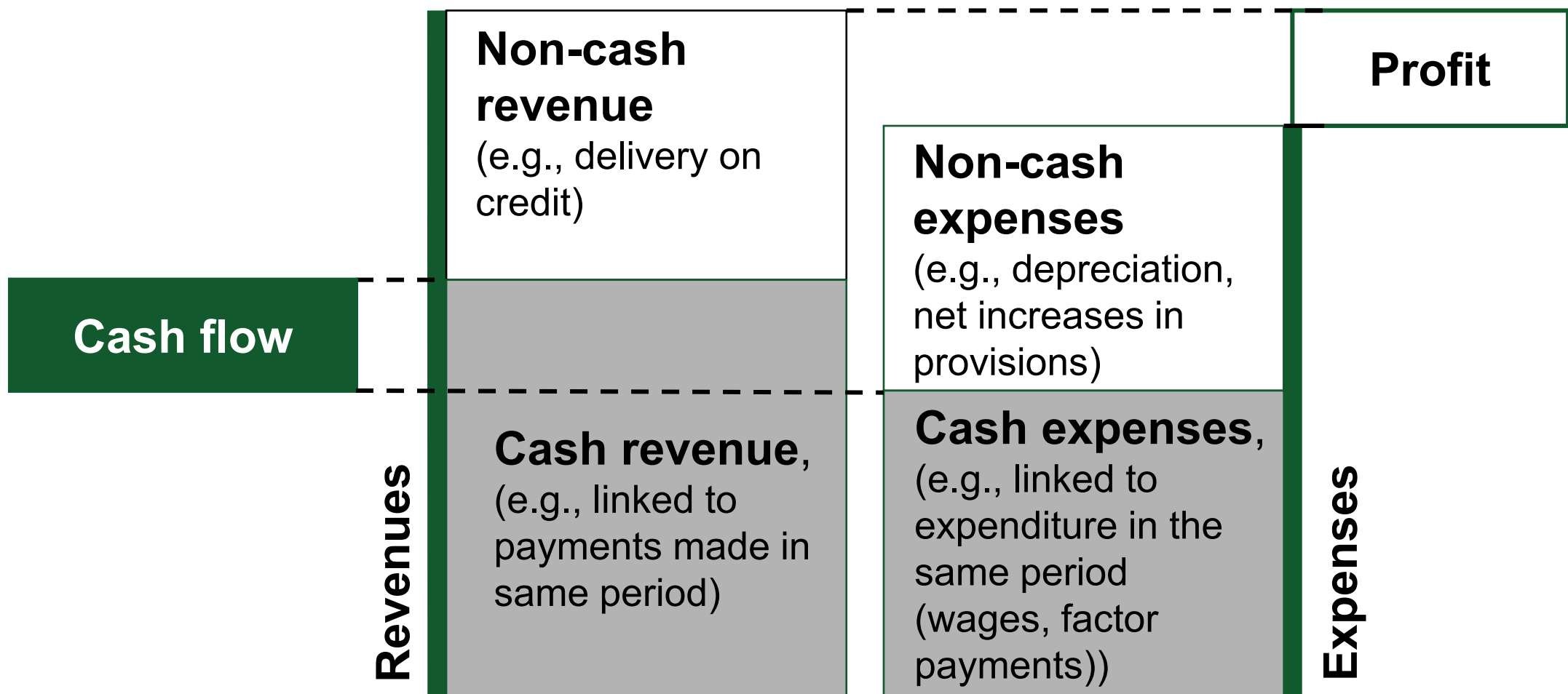
- To account for stock buybacks, the concept of augmented dividends can be applied:

$$\text{Augmented dividend} = \text{dividends} + \text{stock buyback}$$

- The problem with buybacks is that in contrast to dividends, which are smoothed (because of a signaling effect), they often spike.
- Consequence: Buybacks should be normalized by using averages over longer time periods (e.g., five years).
- When using (augmented) dividends, we trust managers to pay out excess cash to shareholders. The large cash balances of many large firms suggest otherwise.

Discounted Cash Flow (DCF) Model

- Instead of using dividends in the Dividend Discount Model (DDM), **free cash flows (FCF)** can also be used to value a company:



DCF Model – Free cash flows as input

- To calculate the free cash flow to the firm, which could be potentially distributed to a firm's creditors and shareholders, the cash flow has to be adjusted for reinvestment needs:

	EBIT	Earnings before interests and taxes
	- Taxes	Taxes on EBIT
Adjustments for Reinvestments	NOPAT	Net operating income after taxes
	± non-cash items	e.g., adding depreciations, deduction of non-cash valuing gains
	-CAPEX	Capital expenditures
	-ΔWC	Change in non-cash working capital
	FCF	Free cash flow to the firm

- Any tax benefits from interest expenses are neglected.

DCF Model – Free cash flows as input (II)

- The reinvestments consist of two parts:
 1. Reinvestment in **long-lived assets**: Difference between **capital expenditures (CAPEX)** and depreciation which is not a cash expense and therefore is added back to net income.
 2. Reinvestment in **short-lived assets**: Change in non-cash **working capital**. Increases in inventory and accounts receivable represent cash tied up in assets that generate no return (i.e., wasting assets). Supplier credit and accounts payable mute the effect on cash flows.

Cash is not accounted for as this is usually invested in low-risk investments such as short-term U.S. government bonds (T-bills).

Estimating horizon values

- The final payment at H is often called **terminal or horizon value**.
- To determine the **horizon value PV_H** , it is often assumed that the free cash flows at the end of the detailed planning period grow uniformly at a constant rate g .
- Hence, the horizon value can be determined as:

$$\begin{aligned} PV_H &= \frac{FCF_{H+1}}{1+r} + \frac{FCF_{H+2}}{(1+r)^2} + \frac{FCF_{H+3}}{(1+r)^3} + \dots = FCF_{H+1} \cdot \left(\frac{1}{1+r} + \frac{(1+g)}{(1+r)^2} + \frac{(1+g)^2}{(1+r)^3} + \dots \right) \\ &= \frac{FCF_{H+1}}{r-g} \end{aligned}$$

- Thus the present value of a corporation is:

$$PV_0 = \frac{FCF_1}{1+r} + \frac{FCF_2}{(1+r)^2} + \dots + \frac{FCF_H}{(1+r)^H} + \frac{1}{(1+r)^H} \times \frac{FCF_{H+1}}{r-g}$$

Discounted Cash Flow Method – VAT Group example

- Analysts predict the following free cash flows to equity per share for VAT Group: $FCF_{2022} = 7.80$, $FCF_{2023} = 8.27$, $FCF_{2024} = 8.37$
- We assume that the free cash flows in the following two years are: $FCF_{2025} = 8.67$, $FCF_{2026} = 8.98$.
- As of the fifth year, the growth of $FCFs$ is forecasted at a constant growth rate of 2% p.a. and the estimated discount rate is 4.02%.
- What is the expected share price of VAT Group?

Discounted Cash Flow Method – Advantages

Advantages of the DCF method:

- The DCF method considers the **cash flows** which were generated from **business activities** in one year.
- Amortization policy and different accounting rules have a significant influence on profits, but not on cash flows. Hence, DCF is **independent from profit considerations and accounting policies**.
- DCF is **independent of dividend payments**, which primarily reflect a decision on the distribution of profits.

Discounted Cash Flow Method – Pitfalls

Potential **pitfalls** of the DCF method:

- Transparency: What level of budgeting / amount of **CAPEX** has been chosen (and why)?
 - Often made assumption: Investments in fixed assets and working capital grow in proportion to sales
- Material correctness: Are the cash flow **forecasts realistic** based on both the company's profile and the economic outlook for the industry and economy at large?
 - Forecast annual sales usually by estimating growth rates.
 - Forecast annual costs which should reflect the competitive environment (marketing, competitive prices, etc.)

Multiple Valuation

- Often it is more informative to compare the relative value (or implied growth rate) of different firms.
- By comparing firms, market expectations can be included in the valuation.
- Procedure:
 1. Determine a peer group.
 2. Choose one (or several) financial ratios.
 3. Calculate the ratios for both the firm to be compared and the peers.
 4. Calculate the mean or median of the peer group as a benchmark.

Multiple Valuation – Primary fields of application

Equity valuation

- Often used as a second / additional valuation method.
- Sometimes as a primary valuation method (e.g., IPOs, minority stakes, private firms, divisions of listed firms, firms that do not generate positive earnings or cash flows, firms in certain industries and emerging markets, young firms).

Investment decisions

- Primary decision tool for style investing (i.e., value / growth investing).
- Frequently used for stock screening.
- The aggregate of such valuation ratios is often used as a measure of market valuation and investor sentiment.

Multiple Valuation – Important Multiples

- Price-earnings multiple (P/E):

$$P/E = \frac{\text{Price}}{\text{Earnings}}$$

- Earnings yield (EY) or Earnings-price multiple (inverse of the P/E):

$$EY = \frac{\text{Earnings}}{\text{Price}}$$

- Dividend yield (D/P):

$$D_1/P_0 = \frac{\text{Dividend in } t = 1}{\text{Price in } t = 0}$$

Multiple Valuation – Important Multiples (II)

- Market-to-book ratio (M/B):

$$M/B = \frac{\text{Share price}}{\text{Book value per share}}$$

- Price-sales ratio (P/S):

$$P/S = \frac{\text{Share price}}{\text{Annual sales per share}}$$

- Enterprise value- $EBITDA$ multiple ($EV/EBITDA$):

$$EV/EBITDA = \frac{\text{Enterprise value}}{EBITDA}$$

EV : Market value of equity +
book value of debt – cash

$EBITDA$: Earnings before interest, taxes,
depreciation, and amortization

Multiple Valuation – Example

- Is Andres Wine undervalued?

Company Name	Trailing PE	Expected Growth	Standard Deviation	PEG
Andres Wine Ltd.	8.96	3.50%	24.70%	2.56
Anheuser Busch	24.31	11.00%	22.92%	2.21
Boston Beer	10.59	17.13%	39.58%	0.62
Brown-Forman	10.07	11.50%	29.43%	0.88
Chalone Wine Group Ltd.	21.76	14.00%	24.08%	1.55
Coca-Cola	44.33	19.00%	35.51%	2.33
Coca-Cola Bottling	29.18	9.50%	20.58%	3.07
Coca-Cola Enterprises	37.14	27.00%	51.34%	1.38
Coors (Adolph)	23.02	10.00%	29.52%	2.30
Corby Distilleries Ltd.	16.24	7.50%	23.66%	2.16
Hansen Natural Corp	9.70	17.00%	62.45%	0.57
Moslon Inc. Ltd.	43.65	15.50%	21.88%	2.82
Mondavi (Robert)	16.47	14.00%	45.84%	1.18
Pepsi Co. Inc.	33.00	10.50%	31.35%	3.14
Todhunter International	8.94	3.00%	25.74%	2.98
Whitman Corp.	25.19	11.50%	44.26%	2.19
Average	22.66	12.60%	33.30%	2.00

Source: Damodaran (2010), based on Value Line

Multiple Valuation – Example (II)

P/E ratio

8.96 vs. 23.57*

Andres Wine seems to be
undervalued.

Problem: How to adjust for Andres Wine's low growth rate?

*23.57 is the peer group average when Andres Wine is excluded.

growth-adjusted P/E ratio (PEG)

$$PEG = \frac{\text{Trailing PE}}{\text{Expected Growth}}$$
$$= \frac{8.96}{3.50} = 2.56$$

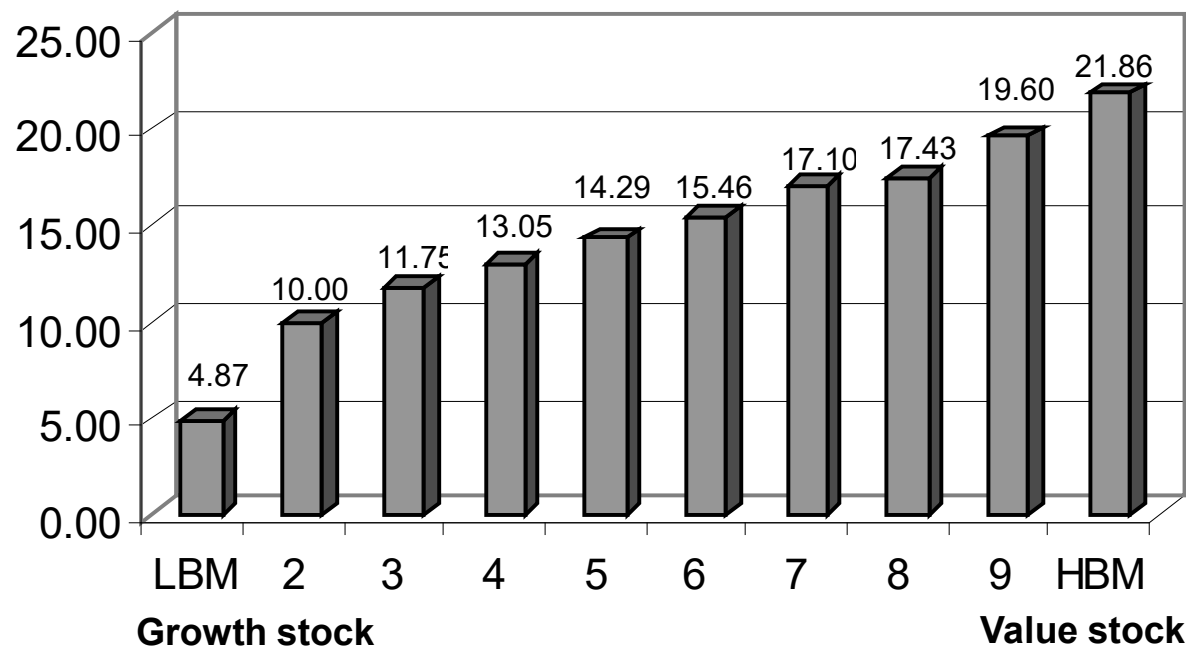
The P/E ratio resulting from the
average PEG and Andres Wine's
expected growth rate is:

$$2.00 \cdot 3.50\% = 7.00$$

Now, with an (implied) adjusted P/E ratio of 7.00, Andres Wine looks overvalued compared to the P/E ratio of 8.96.

Multiple Valuation – Decision tool for investing

- Example: Multiples as decision tool for style investing
- The average annual returns of 10 U.S. stock portfolios sorted based on their book-to-market ratio (1960-2011).



Source: Schmid (2012); data is from Kenneth French's Data Library

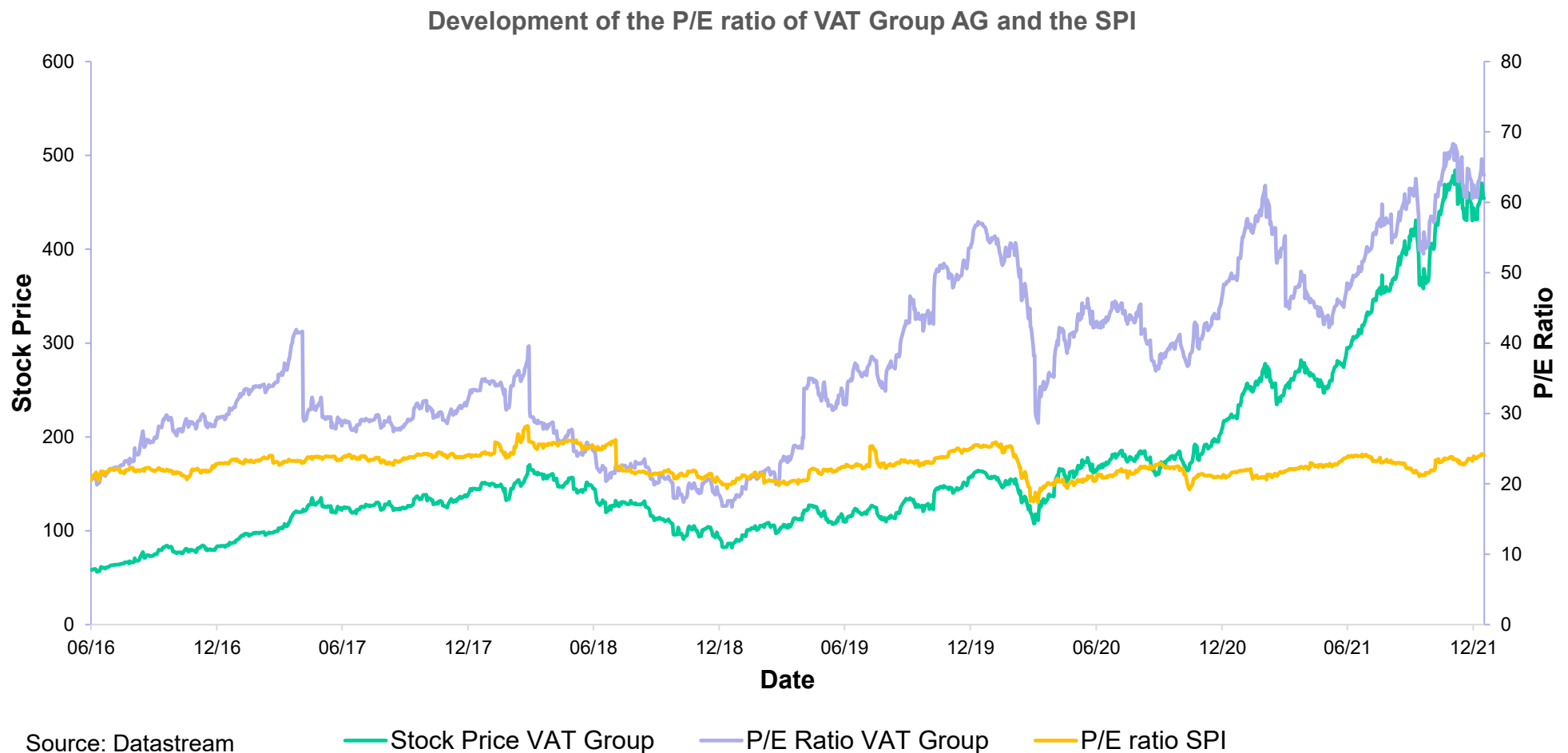
LBM: Low Book-to-market

HBM: High Book-to-market

Multiple Valuation – Example VAT Group

- What is VAT Group's P/E ratio on Dec. 31, 2021? $\rightarrow 454.4 / 7.25 = 62.7$
- What is VAT Group's M/B ratio on Dec. 31, 2021? $\rightarrow 1.36 / 0.6344 = 2.14$
- What do analysts think about VAT Group (as of Dec. 31, 2021)?:
 - How many analysts cover VAT Group (according to their website)?:
 $\rightarrow 14$
 - Consensus EPS forecast (1Y): $\rightarrow$ CHF 4.92
 - Consensus stock price estimate (1Y): $\rightarrow$ CHF 398 (450/270)
 - Consensus stock recommendation: 3.1 (~ hold)
 - Where is the stock price on Dec. 31, 2022?: $\rightarrow$ CHF 252.80

Multiple Valuation – Example VAT Group (II)



Multiple Valuation – Pros and Cons

Pros

- Few assumptions
- Easy to calculate and interpret
- Reflects current market mood (i.e., provides information on market state); comparison between firms at the same time is not affected
- Allows easy comparison between firms and industries

Cons

- Neglect of risk
- Limited forward looking (e.g., D/P)
- Bubble effect (e.g., inflated values in bull market)
- Lack of transparency
 - Choice of peer group
 - Choice of multiples
 - Calculation of multiples

Key takeaways

1. The **present value of a common stock** equals the present value of all future dividends.
2. The **Dividend Discount Model (DDM)** is a simple valuation model which allows to estimate the value of a company by discounting all future dividend payments.
3. The **Gordon Growth Model (GGM)** simplifies the DDM by introducing a (constant) **dividend growth rate**.
4. We can replace the dividends in the DDM by the Free Cash Flows (FCF) to obtain the **Discounted Cash Flow model (DCF)**.
5. By comparing relative valuations across firms, **Multiples** account for market moods and require little assumptions.

Mandatory readings

- Brealey, R., Myers, S., Allen, F., and A. Edmans, 2022, Principles of Corporate Finance, 14th Edition, New York: McGraw-Hill, Chapter 2 (How to Calculate Present Values), Chapter 4 (The Value of Common Stocks), Chapter 5 (Net Present Value and Other Investment Criteria), & Chapter 6 (Making Investment Decisions with the Net Present Value Rule).

Additional readings

- Damodaran, A., 2018, The Dark Side of Valuation: Valuing Young, Distressed, and Complex Businesses, 3rd edition, Pearson.
- Koller, T., Goedhart, G., and D. Wessels, 2020, Valuation: Measuring and Managing the Value of Companies, 7th edition, John Wiley & Sons.
- Fama, E.F., and K.R. French, 2001, Disappearing dividends: Changing firm characteristics or lower propensity to pay?, Journal of Financial Economics 60, 3-43.